COMS 5110 - Homework 3

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Due: March 27, 11:59 PM

GUIDELINES

- Solutions must be typed and submitted as a PDF via Canvas.
- Proofs must be clear and rigorous; do not hand-wave.
- Ensure your file is not corrupted and stays under 5 MB.
- Late policy: No late homework accepted (maximum one late submission up to 12 hours late).

Problem 1. (25 points). A Quantum Compute Fabric Q consists of r intermittent "Entanglement Requests" $E = \{e_1, e_2, \ldots, e_r\}$ and s "Coherence Reservoirs" $R = \{c_1, c_2, \ldots, c_s\}$. The system architecture is subject to the following **Pauli-Exclusion Protocols**:

- Each Coherence Reservoir c_j has a unitary capacity limit; it can stabilize at most one Entanglement Request.
- An Entanglement Request e_i is non-local and atomic; it requires exactly one reservoir and cannot be "sharded" across the fabric.
- Stability is hardware-dependent: for each e_i, there is a specific subset of reservoirs $S_i \subseteq R$ that possess the correct "Phase Signature" to sustain it.

Tasks:

- Quantify the **Saturation Threshold** (the maximal cardinality of stabilized requests).
- Develop an algorithm to determine this threshold, provide a rigorous proof that your procedure reaches the global optimum, and derive its time complexity in terms of r and s.

Proof. We reduce the problem to a maximum flow instance. Consider a directed graph

$$G = (V, E)$$

constructed as follows. The vertex set consists of four parts:

$$V = \{s, t\} \cup E \cup R,$$

where

$$E = \{e_1, \ldots, e_r\}$$

represents the entanglement requests, and

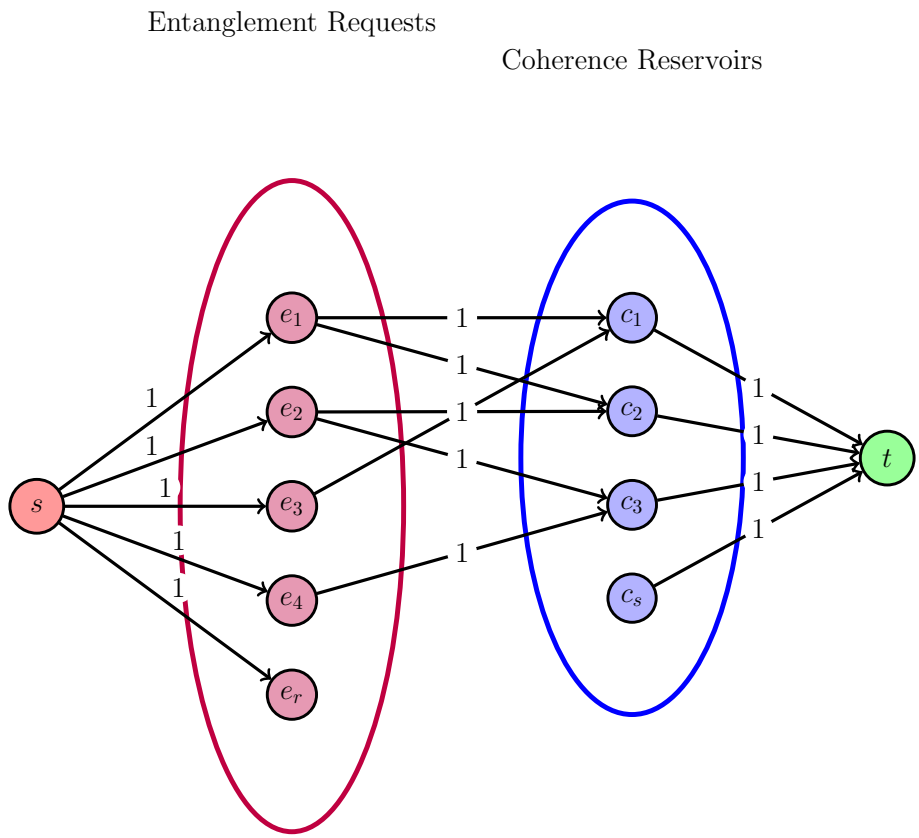
$$R = \{c_1, \ldots, c_s\}$$

represents the coherence reservoirs. The nodes s and t denote the source and sink, respectively.

Edges are added as follows.

- For every entanglement request e_i, add an edge (s, e_i) with capacity 1. This ensures that each request can be stabilized by at most one reservoir.
- For each pair (e_i, c_j) such that $c_j \in S_i$, add an edge (e_i, c_j) with capacity 1. This enforces the hardware-dependency condition: request e_i can only be assigned to a reservoir whose phase signature is compatible with it.
- For every coherence reservoir c_j, add an edge (c_j, t) with capacity 1. This enforces the unitary capacity limit, ensuring that each reservoir stabilizes at most one request.

An example of such a construction is illustrated in the figure below.



We initialize the flow to zero on every edge and run Ford–Fulkerson on the constructed flow network. While there exists an augmenting path from s to t in the residual graph, we augment the flow along that path.

In this network, every augmenting path has the form

$$s \to e_i \to c_j \to t,$$

where $c_j \in S_i$. Thus, each augmentation corresponds to assigning one compatible reservoir c_j to one request e_i. Since all capacities are 1, the bottleneck capacity of every such augmenting path is 1, so each augmentation increases the total flow by exactly 1.

Because there are r edges leaving the source and each has capacity 1, the total capacity leaving s is r. Hence the value of any flow is at most r. Also, since there are s edges entering the sink and each has capacity 1, the total flow is at most s. Therefore, the value of any feasible flow is at most

$$\min(r, s).$$

When the algorithm terminates, the max-flow min-cut theorem implies that the resulting flow is a maximum flow. We now show that its value is exactly the saturation threshold.

A feasible stabilization of k requests corresponds to choosing k distinct pairs

$$(e_i, c_j)$$

such that $c_j \in S_i$, no request appears in more than one pair, and no reservoir appears in more than one pair. From such a stabilization, we construct a flow of value k by sending one unit of flow along

each path

$$s \to e_i \to c_j \to t$$

corresponding to a chosen pair. The capacity constraints are satisfied because each request and each reservoir is used at most once.

Conversely, because all capacities are integers, Ford–Fulkerson returns an integral maximum flow. Thus every edge carries flow either 0 or 1. The set of edges (e_i, c_j) carrying one unit of flow determines a collection of compatible request-reservoir pairs. Since each edge (s, e_i) and each edge (c_j, t) has capacity 1, no request and no reservoir can appear in more than one such pair. Therefore, these pairs define a feasible stabilization whose size equals the value of the flow.

It follows that the value of the maximum flow is exactly the maximum number of requests that can be simultaneously stabilized. Hence, the saturation threshold is equal to the maximum s-t flow in this network.

To analyze the running time, let

$$m = \sum_{i=1}^{r} |S_i|$$

be the number of compatibility edges between requests and reservoirs. Then the network has

$$|V| = r + s + 2$$

vertices and

$$|E| = r + m + s$$

edges.

Using Ford–Fulkerson, each augmenting path increases the flow by 1, and the maximum flow is at most $\min(r, s)$. If each augmenting path is found in $O(|E|)$ time, then the total running time is

$$O\big(\min(r, s) \cdot (r + m + s)\big).$$

Since $m \leq rs$, this may also be written as

$$O\big(\min(r, s) \cdot (r + s + rs)\big).$$

Therefore, this algorithm computes the saturation threshold and is correct.

To explicitly recover the stabilized request-reservoir assignments, after the algorithm terminates we simply inspect the edges (e_i, c_j). Whenever such an edge carries flow 1, we record that request e_i is assigned to reservoir c_j. Since each request and each reservoir can appear in at most one such pair, this reconstruction is valid and requires $O(m)$ time to inspect all compatibility edges. □

Problem 2. (25 points). In a high-volatility labor market, n "Proprietary Analysts" and n "Quantum-Funds" have established a **Non-Poachable Equilibrium** M^*. In this state, there exists no pair (a, f) such that analyst a and fund f possess a mutual "Arbitrage Incentive" to abandon their current contracts for each other.

(a) **Incentive Up-Shifting:** Suppose analyst a currently experiences a "Satisfaction Deficit" (assigned to fund $f = M^*(a)$), which lowers their preference. If a undergoes a "Psychological Re-indexing" where they move their assigned fund f to a superior rank in their internal valuation list, must the equilibrium M^* remain non-poachable? Either provide a formal proof that the equilibrium is still "Non-Poachable" or give a counterexample.

(b) **Incentive Down-Shifting:** Conversely, assume analyst a performs a "Valuation Write-down," moving their current fund f to a subordinate rank in their preference list (while still maintaining f as a valid contractual partner). Does this shift introduce a "Poaching Risk" into the existing M^*? Either provide a formal proof that the equilibrium is still "Non-Poachable" or give a counterexample.

Proof. (a) Let M^* be a stable matching under the original preference profile. Suppose analyst a changes only their own preference list by moving their assigned fund

$$f = M^*(a)$$

to a higher position, while keeping the relative order of all other funds unchanged. We show that M^* remains stable under the new preference profile.

Assume for contradiction that M^* is no longer stable after this change. Then there exists a blocking pair (a', f') under the new preference profile. By definition, this means

$$f' \succ_{a'} M^*(a') \qquad \text{and} \qquad a' \succ_{f'} M^*(f').$$

We consider two cases.

Case 1: $a' \neq a$.

Then analyst a''s preference list did not change at all. Also, no fund's preference list changed. Hence the two inequalities

$$f' \succ_{a'} M^*(a') \qquad \text{and} \qquad a' \succ_{f'} M^*(f')$$

were already true before the modification. Therefore (a', f') was already a blocking pair in the original instance, contradicting the stability of M^*.

Case 2: $a' = a$.

Then the blocking pair is of the form (a, f'), and under the new preference profile we have

$$f' \succ_a M^*(a) = f.$$

But the only change to a's preference list was that $f = M^*(a)$ was moved *upward*. Therefore, if a prefers f' to f after the change, then a must also have preferred f' to f before the change. In other words,

$$f' \succ_a M^*(a)$$

was already true in the original preference profile as well.

Since no fund preferences changed, we also still have

$$a \succ_{f'} M^*(f').$$

Thus (a, f') was already a blocking pair before the modification, again contradicting the stability of M^*.

In both cases we obtain a contradiction. Therefore no blocking pair can arise after moving $M^*(a)$ upward in analyst a's preference list. Hence M^* remains stable.

(b) We construct a simple instance with two analysts $A = \{a_1, a_2\}$ and two funds $F = \{f_1, f_2\}$.

Let the initial preferences be:

$$a_1 : f_1 \succ f_2, \qquad f_1 : a_2 \succ a_1,$$
$$a_2 : f_2 \succ f_1, \qquad f_2 : a_1 \succ a_2.$$

Consider the matching

$$M^* = \{(a_1, f_1), (a_2, f_2)\}.$$

We first verify that M^* is stable. For any unmatched pair (a, f), at least one of the two agents does not prefer the other over their assigned partner. In particular:

- a_1 prefers f_1 over f_2, so (a_1, f_2) is not a blocking pair.
- a_2 prefers f_2 over f_1, so (a_2, f_1) is not a blocking pair.

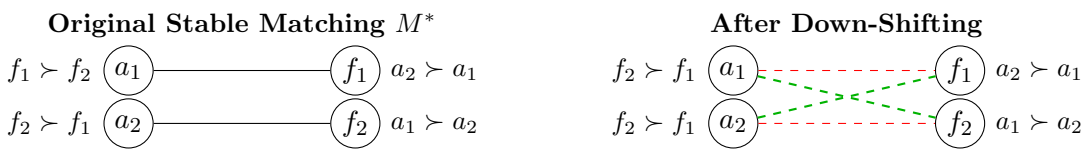
Thus, no blocking pair exists and M^* is a non-poachable equilibrium.

Now suppose analyst a_1 performs a valuation write-down (down-shifting) of their assigned fund f_1, resulting in the updated preference:

$$a_1 : f_2 \succ f_1.$$

All other preferences remain unchanged. Under this modified preference profile, consider the pair (a_1, f_2). We have:

$$f_2 \succ_{a_1} f_1 = M^*(a_1) \quad \text{and} \quad a_1 \succ_{f_2} a_2 = M^*(f_2).$$



(a_1, f_2) blocks; red: unstable, green: new stable

Hence, both a_1 and f_2 strictly prefer each other over their assigned partners, so (a_1, f_2) forms a blocking pair.

Therefore, the matching M^* is no longer stable after the down-shifting operation. This demonstrates that a valuation write-down can introduce a poaching risk into a previously non-poachable equilibrium.

□

Collaboration. For this homework, I have worked solely on the questions without using any resources or help. I have brushed up my answers with OpenAI ChatGPT, but all of the ideas are mine.